

Beyond the Weakest Link: The Skepticism Paradox Why Phishing Works Despite Rising Skepticism

*A Mixed-Methods Study of Loneliness, Social Engineering, and Digital
Trust in Norway*

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Working full thesis draft. Based on [original Phase I](#) (unchanged copy) with Phase 2 survey and Phase 3 interview strands appended. APA 7 (Kildekompasset). Replace all [INSERT] before submission.

Preface

This thesis combines three ideas. First, phishing is not explained by the "weakest link" narrative. Second, general skepticism toward strangers is rising-yet phishing still works. That is the *skepticism paradox*: caution and vulnerability can coexist. Third, socially credible lures may exploit unmet needs for belonging and trust, not ignorance alone.

I thank [supervisor name] for supervision, and OsloMet for academic support during ACIT5910 Phase I.

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Abstract

Purpose and research question. This thesis moves *beyond the weakest link* by explaining why phishing works *despite rising skepticism*. The *skepticism paradox* captures this tension: general caution toward strangers may coexist with vulnerability to socially or institutionally credible lures. The main research question asks which emotional, cognitive, and contextual factors explain this pattern in Norway.

Scope. The study focuses on adults in Norway (with primary data collection among higher-education students and staff), covering phishing, smishing, vishing, and impersonation scams. Malware reverse engineering and deep technical forensics are out of scope.

Methodology. An exploratory sequential mixed-methods design is planned: Phase I establishes theory and instruments; Phase II collects survey data (120-180 respondents); Phase III conducts 12-18 semi-structured interviews and integrates findings through thematic analysis (braun2006thematic).

Expected results. The thesis expects to test four hypotheses on loneliness, lure-specific skepticism, situational pressure, and security culture. Preliminary literature suggests that skepticism protects against stranger-framed scams but fails against lures that imitate belonging, authority, or institutional trust.

Conclusions and suggestions. Phase I concludes that phishing training must move beyond blaming users, address lure-specific impersonation, and connect security awareness with social and emotional context.

Keywords: phishing, loneliness, social engineering, human factors, protection motivation, cybersecurity behaviour

Introduction

Beyond the weakest link

Cybersecurity discourse often treats humans as the weakest link: the predictable failure point in an otherwise robust technical system. Security awareness campaigns, incident reports, and media coverage frequently imply that phishing persists because users lack knowledge, attention, or discipline. This framing is simple and memorable, but it is incomplete.

The weakest-link model assumes that more skepticism uniformly reduces risk. It also tends to locate responsibility at the individual while underplaying how attackers craft messages that exploit trust, urgency, belonging, and institutional legitimacy (hadnagy2018social,conteh2016cybersecurity). When a person clicks, the story becomes “user error.” When a filter fails, the story becomes “system failure.” That asymmetry matters ethically and analytically.

This thesis proposes a different starting point. Phishing is a human *decision* problem disguised as an email. People may know the rules yet still respond unsafely when a message feels meaningful, urgent, or socially rewarding (parker2020contributing,tsai2021phishing). The problem is not only ignorance. It is the interaction between emotional need, situational pressure, message framing, and organisational culture.

The central contribution is the *skepticism paradox* (Section~): phishing works despite rising skepticism because it often does not look like an attack from a stranger. It imitates what people still long for or fear losing—belonging, safety, order, status, help, and control.

Background and relevance

Every year, organisations invest in filters, training, policies, and incident response. Yet phishing and related social engineering attacks remain among the most effective

ways to gain initial access to systems, accounts, and sensitive information (enisa2024threat,cisaPhishing). The problem persists not only because attackers improve their tools, but because they improve their understanding of human motivation (hadnagy2018social).

In Norway, public awareness of digital fraud has increased. People are told to be skeptical of unknown senders, unexpected links, and requests for credentials-skepticism is rising. At the same time, loneliness and social disconnection are widely discussed as social problems across age groups (ssb2025loneliness,helsedirektoratet2024ensomhet). Yet phishing and social engineering remain among the most reported forms of digital fraud (nsm2026risk,nkomSvindel,finansnorge2026svindel). This gap between rising caution and persistent harm motivates the skepticism paradox.

In a time of both rising loneliness and rising general skepticism, phishing becomes dangerous precisely because it does not look like an attack from a stranger. It imitates what people still long for or fear losing: belonging, safety, order, status, help, and control. The thesis therefore asks not only why people click, but why emotionally and socially meaningful lures can bypass skepticism that would otherwise protect them from obvious fraud.

For a Norwegian master thesis in cybersecurity, this topic is timely and feasible. It connects technical security with human behaviour, privacy, trust, and social context without requiring access to classified infrastructure data. It also responds directly to public warnings from Finans Norge, NSM, and Nkom that social engineering remains a live and evolving threat (finansnorge2026svindel,nsm2026risk,nkomSvindel).

Research problem

Main research question

How can phishing remain effective *despite rising skepticism*, and which emotional, cognitive, and contextual factors make individuals especially vulnerable-beyond the

explanation that users are simply the weakest link?

Sub-questions

1. How do loneliness, social connection needs, and general online skepticism relate to phishing susceptibility?
2. Which emotional and situational factors (urgency, authority, fear, shame, stress) most strongly influence unsafe responses?
3. What is the gap between knowing about phishing and acting safely in the moment of decision?
4. Which protective factors-training, culture, verification routines, peer support-are experienced as most effective?

The skepticism paradox

This thesis names and examines a specific tension: the *skepticism paradox*. People in Norway are repeatedly told to distrust unknown links, unexpected requests, and unfamiliar senders (finansnorge2026svindel,nettvettPhishing). At the same time, loneliness signals unmet social connection needs that can increase openness to contact, reassurance, and perceived opportunity (cacioppo2008loneliness,nicholson2021loneliness).

People may report high skepticism toward unknown senders yet remain vulnerable to messages that appear official, socially relevant, or emotionally important (kluetsch2024friend,parker2020contributing). If training assumes that skepticism is uniformly protective, organisations may overlook the emotional and social conditions under which credible impersonation succeeds. If loneliness is treated only as a welfare issue, institutions may miss its relevance to fraud susceptibility.

Scope

- **In scope:** phishing, smishing, vishing, impersonation scams, emotional and social manipulation online.

- **Population:** adults in Norway; primary recruitment among students and staff in higher education (18-35 years emphasis).
- **Out of scope as main focus:** malware reverse engineering, DDoS, ransomware operations, deep technical forensics.
- **Geographic frame:** Norway, with international literature as theoretical backdrop.

Contribution

The thesis aims to contribute academically and practically:

- **Theoretical:** a model of phishing vulnerability that moves beyond the weakest-link narrative, integrating loneliness, lure-specific skepticism, situational pressure, and security culture (rogers1975pmt,floyd2000pmt).
- **Empirical:** mixed-methods evidence on the skepticism paradox, using validated loneliness measures, scenario vignettes, and semi-structured interviews (braun2006thematic).
- **Practical:** recommendations for lure-specific training, non-punitive reporting culture, and integration of student welfare with security awareness (chen2024roleplaying,oslomet2024lunsjvonn).

Norwegian studies by (tjostheim2020phishing,tjostheim2022crt) show that cognitive style predicts phishing susceptibility in representative samples. This thesis examines whether emotional and social variables add explanatory power beyond cognitive style alone.

Thesis structure

This Phase I submission (ACIT5910) comprises the introduction, background, literature review, theoretical framework, methodology, and research plan. Chapters~ - outline the planned analysis and discussion for Phase II (ACIT5920) and Phase III (ACIT5930). Chapter~ presents preliminary conclusions and recommendations.

Background: Phishing as an Entry Point

What is phishing?

Phishing is a social engineering attack in which the victim is manipulated into clicking a malicious link, opening an attachment, revealing credentials, approving an action, or transferring money. The attack succeeds when the message appears credible, relevant, or emotionally meaningful

(hadnagy2018social,conteh2016cybersecurity).

Phishing as the starting point in attack chains

Many serious incidents begin with a human action: a click, a reply, a login, or an approval. From there, attackers may pursue account takeover, data theft, business email compromise, or further exploitation. This makes phishing analytically important even when the final harm is ransomware or data exfiltration

(enisa2024threat,microsoft2024digitaldefense).

Norway and digital trust

Norway is highly digitalised. Citizens rely on BankID, ID-porten, Altinn, digital mailboxes, and online banking. High trust in institutions can be a societal strength, but it may also create fertile ground for impersonation attacks when combined with urgency and fear (politiet2024nettsvindel,nsm2026risk).

AI and the new scale

AI can improve grammar, personalisation, translation, and impersonation (halcyon2026ai,trendmicro2026phishing). That lowers the cost of credible phishing, but it does not replace the need for emotional leverage. AI makes the lure easier to produce; human needs determine whether it works.

Norwegian context: loneliness, students, and phishing

Norway combines high digital trust with measurable social disconnection in key groups. Statistics Norway's quality-of-life survey finds elevated loneliness among young adults (ssb2025loneliness,fhi2024livskvalitet). At the same time, phishing remains the dominant form of digital fraud in Norway (nsm2026risk,nkomSvindel).

At OsloMet and across Norwegian higher education, student loneliness is not marginal. SHoT 2022 found that 29\% of students often miss someone to be with, and campus presence correlates with lower loneliness (shot2022,khrono2024students). Student welfare organisations such as SiO treat loneliness as a public-health and student-life issue (helsedirektoratet2024ensomhet). This thesis connects that welfare context with cybersecurity behaviour.

Literature Review

The weakest-link narrative and its limits

The weakest-link metaphor has shaped cybersecurity training for decades. It implies that security equals the minimum level of human compliance across a system.

Organisations respond with mandatory courses, simulated phishing, and policy reminders. Yet phishing and social engineering remain leading initial-access vectors (enisa2024threat,microsoft2024digitaldefense,trendmicro2026phishing).

Critics of the metaphor argue that it individualises a structural problem. Attackers do not target "weak users" at random; they craft lures matched to context, role, and emotion (hadnagy2018social). Technical systems also create conditions-high email volume, time pressure, fragmented attention-under which verification is costly (jts2024remote,parker2020contributing). When shame and blame follow mistakes, reporting falls and organisational learning suffers (chen2024roleplaying,baylsmith2024phishing).

This thesis therefore treats the weakest-link frame as a hypothesis to be examined, not as an conclusion. If general skepticism were uniformly protective, phishing rates should fall as awareness rises. They have not (nsm2026risk,nkomSvindel). The skepticism paradox offers an alternative explanation: protective caution may be real, but lure-specific.

Phishing and human factors

Research consistently shows that phishing success depends on persuasion strategies, context, workload, and decision pressure rather than intelligence alone (parker2020contributing,tsai2021phishing,baylsmith2024phishing). Employees and students may know the rules yet still click when messages appear legitimate, urgent, or socially relevant.

(frauenstein2020phishing) demonstrate that personality and information-processing style shape susceptibility on social network sites. Their model suggests that individuals who process social cues quickly and trust familiar platform patterns may overlook subtle fraud indicators. (parker2020contributing) extend this finding by showing that contributing factors in organisational contexts include workload, time pressure, and perceived sender legitimacy-variables that standard technical training often ignores.

(tjostheim2020phishing,tjostheim2022crt) provide Norwegian evidence that cognitive reflection predicts resistance to phishing in representative samples. Participants with stronger analytical thinking styles were less likely to comply with deceptive requests even when messages appeared plausible. These findings suggest that cognitive factors matter, but they do not fully explain why emotionally framed lures succeed against people who consider themselves cautious and who perform well on abstract reasoning tasks.

International threat reports confirm that social engineering remains the dominant initial access vector (enisa2024threat,trendmicro2026phishing,microsoft2024digitaldefense). Attackers invest in understanding human motivation because technical defences have improved faster than human decision environments have adapted (hadnagy2018social,conteh2016cybersecurity). The literature therefore supports a shift from blame-oriented models toward contextual models of vulnerability.

Loneliness, belonging, and fraud

Loneliness is not merely the absence of company. (cacioppo2008loneliness) describe it as a distress signal indicating unmet social connection needs-a state that can increase openness to contact, reassurance, and perceived opportunity. The World Health Organization (ons2023loneliness) now treats social connection and loneliness as global health priorities.

Loneliness has long been associated with fraud vulnerability among older adults (alves2008loneliness,wen2022mechanisms). Recent experimental work suggests loneliness can increase susceptibility to misleading offers in middle-aged and older adults (wen2024loneliness). Mechanisms include increased desire for connection, greater trust in persuasive strangers, and reduced access to social verification.

However, the relationship is not uniform across age groups. Among younger adults, FOMO, social media exposure, impulsivity, and platform-specific cues may matter more (przybylski2013fomo,kluetsch2024friend). (popovac2020online) and related work on online deception suggest that platform familiarity and perceived sender identity interact with emotional state. Research gap: loneliness is well studied in older fraud victims, but less systematically examined in Norwegian higher-education contexts and in relation to the skepticism paradox.

The skepticism paradox in practice

People may report high skepticism toward unknown senders yet remain vulnerable to messages that appear official, socially relevant, or emotionally important. Young adults are highly exposed on social platforms and may respond differently to phishing that mimics peers, events, or opportunities rather than strangers (kluetsch2024friend,parker2020contributing).

Public campaigns emphasise distrust of strangers while phishing increasingly impersonates institutions, employers, classmates, and services people depend on daily (finansnorge2026svindel,nettvettPhishing). This creates the paradox named in Section~ : general caution and specific vulnerability can coexist.

This is the empirical face of the skepticism paradox: knowing that fraud exists does not guarantee safe action when a message feels meaningful, urgent, or socially rewarding. Technical indicators (mismatched domains, suspicious headers) may be ignored when the narrative fits an existing need (tsai2021phishing,parker2020contributing). Training that treats all phishing as equally obvious therefore risks a false sense of security among people who would never click

a crude stranger scam yet remain exposed to sophisticated institutional impersonation (tjostheim2022crt).

Protection Motivation Theory

Protection Motivation Theory (PMT) explains protective behaviour through threat appraisal (severity and vulnerability) and coping appraisal (response efficacy and self-efficacy) (rogers1975pmt,floyd2000pmt). In cybersecurity, PMT has been applied to explain compliance with security policies and responses to organisational phishing warnings (baylsmith2024phishing).

For this thesis, PMT is useful but incomplete. It explains motivation to protect oneself, yet phishing often succeeds when the message does not initially look like a threat. Emotional needs may reframe the message as an opportunity rather than a risk. The thesis therefore uses PMT primarily to analyse verification habits, reporting behaviour, and perceived organisational support-coping appraisal factors-rather than to assume that fear-based training alone will prevent clicks (chen2024roleplaying).

Remote work, cognitive bias, and shame

Remote and hybrid study/work environments may reduce informal verification ("Did you really send this?") and increase reliance on digital cues alone (jts2024remote).

When peers and supervisors are not physically present, impersonation becomes harder to detect through quick social confirmation. Students may receive fraudulent messages about exams, fees, or housing at times when administrative communication is already high-volume and stressful.

Shame and fear of blame after clicking can suppress reporting (chen2024roleplaying). Training that relies on simulated phishing without support-seeking components may increase awareness but not safe behaviour under stress. Organisations that punish employees or students who report mistakes create incentives to hide near misses, reducing organisational learning

(baylsmith2024phishing). A non-punitive security culture is therefore treated as a protective factor alongside technical controls.

Cognitive biases relevant to phishing include authority bias (trusting apparent officials), scarcity/urgency effects (acting before verifying), and optimism bias (believing that fraud happens to others). These biases interact with emotional states: loneliness may increase the salience of socially rewarding messages, while stress may reduce the cognitive resources available for verification (cacioppo2008loneliness,tsai2021phishing).

Theoretical synthesis and original contribution

The literature supports a combined view: phishing susceptibility emerges from the interaction of emotional need, situational pressure, message framing, and organisational culture-not from individual ignorance alone. The thesis proposes a combined model centred on the skepticism paradox:

Phishing susceptibility = emotional relevance of lure + situational pressure + unmet social need $\times$ verification opportunity $\times$ non-punitive security culture.

Norwegian national studies complement international findings: (tjostheim2020phishing,tjostheim2022crt) demonstrate that cognitive reflection and data-sharing willingness predict phishing compliance in representative samples. This thesis asks whether loneliness and lure-specific skepticism add explanatory power beyond cognitive style, especially among higher-education adults who are digitally literate yet socially pressured (kluetsch2024friend,shot2022).

Theoretical Framework and Hypotheses

The skepticism paradox as analytical lens

The theoretical contribution of this thesis is to treat phishing not primarily as a failure of knowledge, but as a collision between two simultaneous social trends: rising general skepticism toward strangers and rising subjective loneliness or social need (ssb2025loneliness,khrono2024students,wired2024loneliness). This collision produces the skepticism paradox-the central concept named in the title.

Formally, the paradox can be stated as follows:

Skepticism paradox. Individuals may exhibit high general skepticism toward unknown or unsolicited digital contact while simultaneously remaining vulnerable to phishing that imitates social belonging, institutional authority, or emotionally meaningful opportunity.

The paradox explains why the weakest-link narrative fails empirically and ethically. Empirically, it predicts lure-specific protection: stranger-framed scams should correlate with skepticism; institutional and social lures should not. Ethically, it shifts attention from blaming users to understanding how credible impersonation exploits legitimate human needs (cacioppo2008loneliness,kluetsch2024friend).

The skepticism paradox also reframes loneliness. Loneliness is not introduced as a medical diagnosis in this thesis, but as a signal of unmet social connection that may increase the salience of belonging-framed lures (S2, S6). Age will be treated as a control and exploratory variable because prior research suggests loneliness may operate differently across cohorts (wen2024loneliness,barstad2021ensomme).

Hypotheses

H1

Higher loneliness is associated with higher self-reported phishing susceptibility, especially for socially framed lures (S2, S6).

H2

Higher general skepticism reduces susceptibility for stranger-framed lures but not for institutionally or emotionally relevant lures (interaction effect).

H3

Situational pressure (urgency, authority, stress) partially mediates the relationship between loneliness and unsafe intended behaviour.

H4

Participants with stronger verification habits and supportive, non-punitive security culture report lower susceptibility (PMT coping appraisal).

Methodology

Design

Recommended design: exploratory sequential mixed methods (braun2006thematic).

Phase I (this document) establishes theory and instruments. Phase II collects survey data. Phase III conducts interviews and integrates findings.

1. **Phase 1 (ACIT5910):** literature review, theory, instruments, ethics plan.
2. **Phase 2 (ACIT5920):** online survey (quantitative + short open text).
3. **Phase 3 (ACIT5930):** semi-structured interviews; thematic analysis integrated with survey findings.

Population and sample

- **Population:** students and staff in higher education in Norway.
- **Target sample:** 120-180 survey responses; 12-18 interviews.
- **Inclusion:** adults 18+ with regular digital communication for study/work.
- **Recruitment:** OsloMet programmes, student networks, LinkedIn, privacy-preserving invitation text.

Before main data collection, a pilot with 10-15 OsloMet students/staff will test completion time (target under 12 minutes), clarity of scenario vignettes, floor/ceiling effects on loneliness items, and whether skepticism items discriminate.

Instruments

The survey will combine validated and adapted instruments:

- **Loneliness:** short-form items adapted from established scales measuring subjective social disconnection (3-6 items), aligned with Norwegian population research (ssb2025loneliness,nicholson2021loneliness).

- **General online skepticism:** items measuring distrust of unknown senders, links, and unsolicited requests.
- **Scenario vignettes (S1-S6):** see Section~ .
- **Situational pressure:** perceived urgency, authority, stress, and time pressure at time of response.
- **Verification habits:** frequency of checking sender identity, using separate channels, and consulting peers.
- **Security culture:** perceived blame, support for reporting, and quality of institutional communication (baylsmith2024phishing,chen2024roleplaying).
- **Demographics and controls:** age, study level, digital literacy self-rating, prior phishing training exposure.

Open-text items will capture qualitative nuance: descriptions of near misses, reasons for hesitation, and perceived gaps between policy and practice.

Interviews will explore:

- what the message looked like and why it felt credible;
- emotional state at the time (lonely, stressed, busy, afraid);
- whether they knew the advice but still hesitated or clicked;
- experiences of training, shame, and reporting culture.

Scenario vignettes (S1-S6)

Six short vignettes will manipulate lure framing:

1. **S1 Stranger:** generic prize or account verification from unknown sender.
2. **S2 Social:** message mimicking a peer, study group, or campus event.
3. **S3 Institutional:** BankID, Altinn, or university IT support impersonation.
4. **S4 Authority:** urgent request from apparent manager or exam office.

5. **S5 Opportunity:** internship, job, or exclusive offer with time pressure.

6. **S6 Belonging:** invitation to community, support, or social connection.

Respondents indicate intended behaviour (click, verify, ignore, report) on a fixed scale. H1 and H2 specifically compare socially and institutionally framed lures against stranger-framed baselines.

Ethics

The study will require NSD approval and informed consent before data collection. Data will be pseudonymised; no live deceptive phishing against participants will be conducted without explicit ethical approval. Participants may withdraw without consequence. Special care applies when asking about fraud victimisation or shame, as these topics may trigger distress (alves2008loneliness).

Data handling will follow GDPR principles: survey hosted on an institution-approved platform; data stored encrypted on OsloMet-approved storage; access limited to researcher and supervisor; retention aligned with NSD approval; deletion after the thesis archive period. Interview audio will be transcribed and direct identifiers removed before analysis.

Limitations

- Self-report bias and social desirability may understate unsafe intentions.
- Convenience sampling may overrepresent cybersecurity-aware students.
- Loneliness scales measure subjective experience, not objective isolation.
- Cross-sectional survey data cannot establish full causality.
- (wen2024loneliness) found weaker loneliness-fraud effects in younger adults; H1 may require subgroup analysis.

Analysis plan

Survey data: descriptive statistics, correlation, regression, and planned moderation/mediation tests for H1-H4. Interview data: reflexive thematic analysis (braun2006thematic). Integration: joint display of quantitative patterns and qualitative mechanisms in Phase III.

Survey Methodology (Phase 2)

Research design

The study uses an exploratory sequential mixed-methods design (Braun & Clarke, 2006). Phase 2 implements the quantitative strand: a cross-sectional online survey with scenario vignettes and validated/adapted scales. Phase 3 will add semi-structured interviews and integrate findings. The quantitative phase precedes interviews so that interview sampling can target variation in loneliness scores and vignette responses.

The design prioritises ethical safety: vignettes present hypothetical messages; no live deceptive phishing is deployed against participants. Intended behaviour is measured on a fixed ordinal scale rather than actual clicks on malicious links.

Within-subject comparison across six vignettes increases statistical power relative to between-subject designs and mirrors the real-world condition in which the same individual receives multiple lure types over time. Each participant acts as their own control for general scepticism and loneliness, while lure framing varies systematically.

The quantitative phase answers whether self-reported traits and contexts predict intended unsafe behaviour across lure types. It does not measure actual victimisation rates or organisational incident data; those would require different methods and permissions beyond this thesis scope.

Conceptual model and variables

Figure 1 (conceptual, from Phase 1) maps the scepticism paradox for empirical testing. Independent variables: loneliness (continuous), general online scepticism (continuous), situational pressure (continuous/mediator). Within-subject factor: lure type (six vignettes). Dependent variable: intended unsafe behaviour (click immediately vs verify/ignore/report). Moderators/mediators for H3-H4: situational pressure, verification habits, security culture.

Control variables reduce confounding: age, role (student/staff), prior phishing training exposure, self-rated digital literacy. Optional cognitive reflection item enables comparison with Tjosheim and Waterworth (2020, 2022) Norwegian samples.

The model treats phishing susceptibility as lure-specific rather than trait-only. A respondent may score high on scepticism and low on loneliness yet still select "click" on S3 if institutional authority cues dominate—a pattern that would support H2 and contradict uniform weakest-link assumptions.

Pilot study

Before main data collection, a pilot with 10-15 students and staff at OsloMet tested instrument clarity, completion time, and floor/ceiling effects. Pilot criteria: median completion under 12 minutes; no item with more than 5% missing; vignette instructions understood without assistance; loneliness and scepticism items show sufficient variance.

Pilot revisions expected: shorten redundant items, clarify Norwegian/English terminology for institutional lures (BankID, Altinn, IT-support), and adjust vignette ordering to reduce carry-over effects. Pilot data are excluded from main hypothesis tests but inform instrument finalisation documented in Appendix A.

[INSERT: Pilot N, completion time median, items revised, date conducted]

Population, sampling, and recruitment

Population: adults aged 18+ in Norwegian higher education with regular digital communication for study or work. Target sample: 120-180 complete survey responses (power adequate for moderation analysis with continuous predictors; see analysis plan).

Inclusion: student or employee at a Norwegian university or university college; uses email or messaging for study/work weekly; reads Norwegian or English.

Exclusion: under 18; no regular digital communication; completed the survey previously (duplicate IP/cookie check).

Recruitment channels: OsloMet student email lists (with faculty approval), student organisation social media, LinkedIn (higher-education Norway), and snowball sharing with privacy-preserving invitation text. No personal phishing data are collected; recruitment materials state the academic purpose and SIKT approval reference.

[INSERT: Recruitment start/end dates, institutions reached, response rate]

Survey platform and procedure

The survey is hosted on an institution-approved platform (e.g. Nettskjema/Digital Survey Services via OsloMet). Participants access via anonymous link or QR code. Flow: information sheet, informed consent checkbox, demographics, loneliness and scepticism scales, situational pressure and culture items, vignettes S1-S6 in randomised order (to control order effects), verification habits, open-text items, debrief with fraud reporting resources (Finans Norge, NSM Nettvett).

Estimated duration: 10-12 minutes. No compensation planned unless approved by supervisor and SIKT. Participants may withdraw until submission; partial responses are deleted on withdrawal request.

Randomisation of vignette order uses built-in survey logic to reduce priming effects: if S1 (stranger) always appeared first, subsequent institutional lures might appear more credible by contrast. Counterbalancing checks whether order affects mean unsafe intention.

Mobile compatibility is required because many students complete surveys on phones—the same device used for smishing targets. Layout is single-column with minimum 12 pt equivalent text.

Instruments and operationalisation

Loneliness (3-6 items): adapted from established short forms aligned with SSB loneliness indicators (Statistics Norway, 2025). Example stem: "How often do you feel that you lack companionship?" (1 = never to 5 = always). Mean score forms continuous predictor; higher = greater subjective loneliness.

General online scepticism (4-6 items): distrust of unknown senders, unexpected links, unsolicited attachments. Example: "I am cautious about clicking links from people I do not know." (1 = strongly disagree to 5 = strongly agree).

Situational pressure (4 items): perceived urgency, authority, stress, time pressure during survey completion as proxy for decision context.

Verification habits (4 items): frequency of checking sender via separate channel, inspecting URL, consulting peers, delaying response when unsure.

Security culture (4 items): perceived blame after mistakes, comfort reporting near misses, quality of institutional security communication (Bayl-Smith et al., 2024; Chen et al., 2024).

Controls: age, gender (optional), role (student/staff), study level, prior phishing training, self-rated digital literacy, cognitive reflection short item (optional, compare Tjostheim & Waterworth, 2020).

Full item list: Appendix A.

Scenario vignettes (S1-S6)

Six vignettes manipulate lure framing. Each presents a short message in Norwegian context. Response scale per vignette: 1 = Would click/open immediately; 2 = Would verify via separate channel first; 3 = Would ignore; 4 = Would report as suspicious. Primary dependent variable: unsafe intention (1 vs 2-4) and ordinal susceptibility score.

S1 Stranger: SMS from unknown number claiming prize win, link to "claim reward."

S2 Social: Message appearing to be from a fellow student inviting you to a private study group WhatsApp link before exam.

S3 Institutional: Email styled as OsloMet IT support requesting immediate password reset via link due to "security incident."

S4 Authority: Email appearing from exam office demanding fee payment within 2 hours to avoid registration hold.

S5 Opportunity: LinkedIn-style message offering exclusive internship interview slot; link to "confirm availability" within 24 hours.

S6 Belonging: Invitation to a "student wellbeing circle" video call for people "who often feel left out," with warm tone and signup link.

Full vignette texts: Appendix B. Alt-text and plain-language summaries included for accessibility (handbook WCAG requirement).

Ethics and data management

Ethics application submitted to SIKT (Norwegian Agency for Shared Services in Education and Research) before main collection. [INSERT: SIKT reference number, approval date]. Informed consent covers anonymised storage, thesis use, and optional contact for Phase 3 interviews.

GDPR compliance: data stored encrypted on OsloMet-approved systems; access limited to researcher and supervisor; retention per SIKT approval; deletion after archive period. No names, email addresses, or IP addresses stored with response data. Fraud victimisation questions are optional and preceded by content warning.

Debrief provides links to Finans Norge (svindel.no), NSM Nettvett, and student welfare services. Participants experiencing distress may contact SiO or equivalent.

The study avoids deception about study purpose: participants know they are evaluating hypothetical messages for a thesis on phishing and loneliness. This differs from unannounced simulated phishing in organisations, which raises separate ethical

issues (Chen et al., 2024). Transparency may increase social desirability bias toward "verify" responses; this is acknowledged as a limitation.

Data cleaning and quality checks

Exclusion rules: completion time under 3 minutes (insufficient engagement); straight-lining on all Likert items; duplicate responses. Missing data: preregistered maximum 20% missing on scale items; multiple imputation not planned for main analysis given ordinal outcomes; listwise deletion for primary regression models with sensitivity analysis.

Coding: vignette unsafe = 1 if response is "click immediately"; scepticism and loneliness as continuous; lure type as within-subject factor (six levels) in repeated-measures framework.

Analysis plan

Software: R or SPSS. Descriptive statistics for all variables and vignette response distributions. Reliability: Cronbach's alpha for multi-item scales (target alpha > .70).

H1: Mixed-effects or repeated-measures logistic regression with loneliness predicting unsafe intention; interaction term loneliness x lure type; focus on S2 and S6 vs S1.

H2: General scepticism x lure type interaction; contrast stranger (S1) vs institutional/social lures.

H3: Mediation analysis (PROCESS or structural equation) with situational pressure as mediator between loneliness and unsafe intention.

H4: Verification habits and security culture as moderators or predictors of safer responses (verify/report vs click).

Controls in all models: age, role, prior training, digital literacy. Subgroup exploratory analysis: age bands 18-24 vs 25+ given Wen et al. (2024).

Assumptions: check multicollinearity (VIF), residual patterns for linear components; sensitivity analysis with alternative cut-offs for unsafe behaviour.

[INSERT: R/SPSS script reference in appendix when analysis complete]

Sample size rationale: $N = 120-180$ provides adequate power for medium-sized interaction effects in logistic mixed models with six repeated measures per participant. A conservative rule of thumb requires 15-20 observations per predictor in multivariate models; the upper target ($N = 180$) allows listwise deletion without falling below power thresholds.

Multiple comparison correction: Benjamini-Hochberg FDR applied across primary hypothesis tests (H1-H4) to control false discovery rate while retaining power relative to Bonferroni.

Validity, reliability, and limitations

Construct validity: vignettes reviewed by supervisor and pilot participants for face validity in Norwegian HE context. Self-report susceptibility may differ from behaviour; vignettes approximate decision intention under hypothetical conditions (common in phishing research; Parker & Flowerday, 2020).

External validity: convenience sample may overrepresent digitally literate students; findings generalise cautiously beyond higher education.

Limitations stated in Phase 2: cross-sectional design; self-report bias; no causal claims; loneliness measured subjectively; English/Norwegian bilingual items may affect subsets.

Results: Quantitative Findings

Sample and descriptives

This section presents preliminary quantitative results from the Phase 2 survey.

[INSERT when main collection complete: final N, demographics table, recruitment summary.] Table 1 shows the planned descriptive output structure.

Table 1. Sample characteristics (main survey). [INSERT data]

Variable	n	% or M (SD)
Total valid responses	[]	100%
Students	[]	[]%
Staff	[]	[]%
Age 18-24	[]	[]%
Age 25-34	[]	[]%
Prior phishing training (yes)	[]	[]%
Loneliness scale mean	[] ([])	
General scepticism mean	[] ([])	

Vignette response patterns

Figure 1 (planned) displays the percentage selecting "click immediately" per vignette. Based on Phase 1 literature, stranger-framed S1 is expected to show lower unsafe intention among high-scepticism respondents, while S3 (institutional) and S6 (belonging) may show higher unsafe intention regardless of general scepticism—consistent with the scepticism paradox.

[INSERT: Figure 1 bar chart S1-S6 unsafe intention percentages]

Table 2. Intended unsafe behaviour by vignette (click immediately). [INSERT data]

Vignette	Lure type	Click %	Verify %	Ignore %	Report %
S1	Stranger	[]	[]	[]	[]

Vignette	Lure type	Click %	Verify %	Ignore %	Report %
S2	Social	[]	[]	[]	[]
S3	Institutional	[]	[]	[]	[]
S4	Authority	[]	[]	[]	[]
S5	Opportunity	[]	[]	[]	[]
S6	Belonging	[]	[]	[]	[]

Hypothesis testing (preliminary)

H1 (loneliness and social lures): [INSERT: regression coefficient, OR, p-value, interaction loneliness x S2/S6]. Expected direction: positive association between loneliness and unsafe intention for S2 and S6; weaker or null for S1.

H2 (scepticism x lure type): [INSERT: interaction term results]. Expected: scepticism negatively associated with S1 unsafe intention; non-significant or weaker for S3 and S6.

H3 (mediation): [INSERT: indirect effect estimate, CI]. Expected: partial mediation by situational pressure.

H4 (coping appraisal): [INSERT: verification habits and culture predictors]. Expected: higher verify/report rates among those with strong verification habits and non-punitive culture perceptions.

Table 3. Summary of hypothesis tests. [INSERT statistics]

Hypothesis	Model	Key statistic	p	Supported?
H1	Loneliness x lure (S2,S6)	[]	[]	[]
H2	Scepticism x lure type	[]	[]	[]
H3	Mediation (pressure)	[]	[]	[]
H4	Verification + culture	[]	[]	[]

Open-text preliminary themes

Two open items asked about near misses and what makes messages feel credible.

Phase 2 coding is descriptive until Phase 3 thematic analysis. Preliminary codebook:

(1) institutional branding; (2) time pressure; (3) social inclusion; (4) shame about almost clicking; (5) knowledge-behaviour gap.

[INSERT: 2-3 anonymised quote excerpts illustrative of scepticism paradox]

Interview Methodology (Phase 3)

Design and sampling

Phase 3 implements the qualitative strand of the exploratory sequential mixed-methods design (Braun & Clarke, 2006). Interviews follow survey completion so that sampling can target variation identified in Phase 2.

Target sample: 12-18 semi-structured interviews. Purposive sampling from survey opt-in pool, stratified by: (1) loneliness score tertile (high/medium/low); (2) vignette profile (high susceptibility on S3 or S6 vs low); (3) role (student vs staff). Aim for diversity in gender and age where possible without compromising anonymity.

Inclusion: completed Phase 2 survey; consented to follow-up contact; aged 18+; comfortable interviewing in Norwegian or English.

Exclusion: insufficient language proficiency; inability to consent; duplicate participation.

[INSERT: Final interview N, recruitment period, opt-in rate from survey]

Interview guide and procedure

Interviews use a semi-structured guide (Appendix A) with core topics: experiences with suspicious messages; credibility cues; emotional context (loneliness, stress, exam periods); knowledge-behaviour gaps; training and reporting culture; reactions to vignettes S3 and S6 if not discussed spontaneously.

Format: video or audio call (Teams/Zoom) or in-person at OsloMet, 45-60 minutes. Recorded with consent; transcribed verbatim (or intelligent verbatim) by researcher; pseudonyms assigned (P1-P18).

Opening: reconfirm consent, explain confidentiality, emphasise no wrong answers. Warm-up: general digital communication habits. Core blocks: (1) near-miss narratives; (2) institutional vs stranger messages; (3) loneliness and social connection

(sensitive - allow skip); (4) organisational support and shame; (5) debrief and member-check offer.

Member checking: summary sent to willing participants for factual correction (not veto over interpretation).

Qualitative analysis approach

Analysis follows reflexive thematic analysis (Braun & Clarke, 2006; Braun & Clarke, 2021). Six phases: familiarisation with transcripts; initial coding; theme search; theme review; theme definition and naming; report writing.

Coding software: NVivo, Taguette, or manual coding in Word/Excel. Initial codes generated inductively from data and deductively from the scepticism paradox framework (lure-specific credibility, knowledge-behaviour gap, shame/reporting, institutional trust).

Reflexivity journal maintained: researcher assumptions about weakest-link narrative, loneliness stigma, and cybersecurity culture documented to reduce unexamined bias.

Trustworthiness: thick description of quotes; audit trail of codebook revisions; negative case analysis (interviews contradicting H1/H2); peer debrief with supervisor.

Integration with survey data

Mixed-methods integration uses a joint display approach (Fetters et al., 2013): rows represent themes; columns link to survey statistics (e.g. S6 click rate among high-loneliness tertile). Integration questions: Do interviews explain quantitative interaction effects? Do qualitative cases illustrate outliers?

Triangulation protocol: convergence (qual + quant agree), expansion (qual adds mechanism), discordance (qual challenges quant - reported transparently).

[INSERT: Reference to integration table in Results chapter]

Ethics (interview strand)

Interview strand covered under SIKT approval from Phase 2 [INSERT: reference].

Separate interview consent form. Transcripts stored encrypted; identifiers removed; audio deleted after verified transcription per retention schedule.

Sensitive topics: fraud victimisation, loneliness, shame. Participants may skip questions, pause, or withdraw. Contact information for SiO/student welfare provided before and after interview.

Results: Qualitative Themes

Overview of sample

[INSERT: Table of pseudonyms, role, loneliness tertile, vignette profile, interview duration. No identifying details.]

Theme 1: Lure-specific credibility

Expected theme: participants distinguish stranger scams from institutional or social messages. Stranger lures described as "obvious" or "ridiculous"; institutional lures (BankID, IT, exam office) described as "could be real" even by sceptical participants.

[INSERT: 2-4 anonymised quotes supporting Theme 1]

Relation to H2: if supported, quotes should show general scepticism coexisting with institutional trust - the scepticism paradox in participants' own words.

Theme 2: Emotional context and loneliness

Expected theme: belonging and social lures gain credibility during lonely or stressful periods (exam season, relocation, remote study). Not all high-loneliness scorers endorse this; negative cases noted.

[INSERT: 2-4 quotes. Handle with care - avoid implying loneliness causes gullibility.]

Relation to H1: qualitative depth on when S2/S6 feel tempting despite awareness training.

Theme 3: Knowledge-behaviour gap

Expected theme: participants articulate correct advice ("check sender," "use separate channel") yet describe moments of almost clicking due to urgency, multitasking, or emotional pull.

[INSERT: quotes on gap between knowing and doing]

Relation to sub-question 3 from Phase 1 and situational pressure mediation (H3).

Theme 4: Shame, blame, and reporting culture

Expected theme: near misses hidden when culture feels punitive; supportive reporting linked to willingness to ask for help. Aligns with PMT coping appraisal (H4) and Chen et al. (2024).

[INSERT: quotes on reporting comfort or fear of blame]

Theme 5: Training gaps

Expected theme: awareness materials focus on stranger scams; participants request examples of social and institutional impersonation relevant to student life.

[INSERT: quotes on training relevance]

Practical implication for OsloMet and comparable institutions.

Table 1. Thematic overview and link to hypotheses. [INSERT/refine after coding]

Theme	Brief description	Linked hypothesis	Example quote ID
1. Lure-specific credibility	Institutional/social lures feel more credible than stranger scams	H2	[P?]
2. Emotional context	Loneliness/stress increases openness to belonging lures	H1, H3	[P?]
3. Knowledge-behaviour gap	Rules known but not applied under pressure	H3	[P?]
4. Reporting culture	Shame suppresses disclosure; support enables reporting	H4	[P?]
5. Training gaps	Generic training; need lure-specific examples	Practical	[P?]

Integrated Findings

Joint display: quantitative patterns and qualitative mechanisms

This section integrates Phase 2 and Phase 3 findings. [INSERT: Narrative weaving survey statistics with interview excerpts.]

Example integration structure (to complete with real data): If survey shows higher S6 unsafe intention in high-loneliness tertile, interviews should illustrate mechanisms (e.g. desire for connection during isolated exam period). If H2 interaction is significant, interviews should show participants who "never click stranger links" but hesitated on IT reset emails.

Table 2. Integration matrix (joint display). [INSERT data]

Survey finding	Related theme	Illustrative quote	Integration type
[e.g. S6 click % high in lonely tertile]	Theme 2	"[quote]"	Convergence
[e.g. S1 vs S3 scepticism interaction]	Theme 1	"[quote]"	Expansion
[discordant case if any]	[theme]	"[quote]"	Discordance

Answering the research questions

Main RQ: [INSERT synthesized answer drawing on quant + qual]

Sub-Q1 (loneliness, scepticism, susceptibility): [INSERT]

Sub-Q2 (emotional/situational factors): [INSERT]

Sub-Q3 (knowledge-behaviour gap): [INSERT]

Sub-Q4 (protective factors): [INSERT]

Discussion

The scepticism paradox revisited

The discussion interprets integrated findings through the scepticism paradox lens. If convergent evidence shows lure-specific protection, the weakest-link narrative is insufficient as both training model and incident explanation (Hadnagy, 2018).

Norwegian high-trust digital context (BankID, Altinn, OsloMet IT) creates predictable impersonation surfaces. General stranger-scepticism, promoted in public campaigns (Finans Norge, 2026; NSM, 2026), may not transfer to domains where trust is functional and necessary.

[INSERT: Interpretation of whether paradox is supported, partially supported, or requires revision based on actual findings]

Theoretical contributions

The thesis contributes: (1) a named paradox tested with mixed methods in Norwegian higher education; (2) integration of loneliness research with human-factors cybersecurity; (3) extension of PMT toward coping appraisal and reporting culture when threats are not initially recognised as such (Rogers, 1975; Bayl-Smith et al., 2024).

Comparison with Tjostheim and Waterworth (2020, 2022): cognitive reflection remains relevant; this thesis adds emotional and social framing, especially for institutional lures not tested in prior Norwegian samples.

Practical recommendations

Recommendation 1 - Lure-specific training: Replace generic "don't click unknown links" with scenario-based modules covering IT impersonation, exam/fee authority lures, and social/belonging messages (Finans Norge, 2026; Nettvett.no & NSM, n.d.).

Recommendation 2 - Non-punitive reporting: Near-miss reporting channels with explicit no-blame policy; align with Chen et al. (2024) support-seeking training rather than fear-only simulations.

Recommendation 3 - Student welfare integration: Coordinate security communications with student loneliness initiatives (SHoT Study, 2022; OsloMet, 2024) without stigmatising lonely students as "high risk."

Recommendation 4 - Verify-by-default routines: Separate-channel verification for unexpected IT, fee, or BankID requests; institutional templates showing how legitimate messages appear.

Recommendation 5 - High-risk periods: Targeted reminders during exam periods, tax season, and onboarding when authority and stress lures peak.

[INSERT: Tailor recommendations to supported/unsupported hypotheses]

Limitations

Methodological: self-report survey; hypothetical vignettes; convenience sample; cross-sectional design; single country; higher-education focus.

Qualitative: researcher interpretation; social desirability in interviews; opt-in bias among interviewees willing to discuss fraud and loneliness.

Integration: quant and qual samples overlap partially but not all interviewees need represent survey median.

Generalisation: findings apply cautiously to Norwegian HE adults, not all populations or live phishing behaviour.

Future research

Longitudinal study tracking loneliness and near misses over an academic year.

Intervention study comparing lure-specific emotional-aware training vs standard modules.

Cross-Nordic comparison of institutional trust and phishing susceptibility.

Organisational ethnography of reporting culture in university IT and student admin units.

Conclusion

Oral defence preparation (notes)

Presentation structure (~30 min): (1) problem and paradox; (2) methods overview; (3) key results quant + qual; (4) integrated finding example; (5) recommendations; (6) limitations.

[INSERT: 3-5 core slides storyline after results known]

Appendices

Survey and ethics (Phase 2)

Appendix A: Survey items (draft)

L1. I lack companionship. [1-5] | L2. I feel left out. [1-5] | L3. I feel isolated from others. [1-5]

SK1. I am cautious about links from unknown senders. [1-5] | SK2. Unexpected attachments make me suspicious. [1-5] | SK3. I trust messages from official institutions without checking. [reverse, 1-5]

VH1. I verify unexpected requests via a separate channel. [1-5] | VH2. I check URL before clicking. [1-5]

SC1. My institution treats phishing mistakes punitively. [reverse] | SC2. I would feel comfortable reporting a near miss.

Open: Describe a time you almost clicked a suspicious message. What made it credible?

Appendix B: Vignette texts (full draft)

S1 (Stranger): "Gratulerer! Du har vunnet 10 000 kr i varelotteriet. Klikk her for a hente premien innen 24 timer: [link]"

S2 (Social): "Hei! Det er [fornavn] fra DATA2000-gruppen. Vi har lage en WhatsApp-gruppe til eksamen - bli med her sa deler vi oppsummeringer: [link]"

S3 (Institutional): "OsloMet IT: Av sikkerhetsgrunner ma du tilbakestille passordet ditt innen i kveld. Klikk her: [link]. Unnlatelse kan medfore sperret konto."

S4 (Authority): "Eksamenskontoret: Utestaende eksamensgebyr ma betales innen 2 timer for a unnga studiestopp. Betal her: [link]"

S5 (Opportunity): "Hei, jeg representerer [selskap]. Vi har et eksklusivt internship-intervju til deg - bekreft tidspunkt innen 24t: [link]"

S6 (Belonging): "Hei, vi inviterer deg til en liten online gruppe for studenter som ofte foler seg alene. Ingen drommer, bare støtte. Bli med her: [link]"

Appendix C: Consent and debrief (outline)

Consent key points: voluntary participation; anonymised data; thesis and academic publication use; right to withdraw; contact researcher/supervisor; SIKT approval reference.

Debrief: study purpose (scepticism paradox); no real credentials collected; resources - svindel.no, nettvett.no, SiO/student welfare.

Appendix D: Recruitment invitation (draft text)

Subject: Invitation to research survey - digital security and social connection (10 min)

Body: You are invited to participate in an anonymous OsloMet master thesis survey on phishing awareness and social connection among students and staff. The study uses hypothetical message scenarios—not real phishing. Approved by SIKT [reference]. Link: [URL]. Contact: [researcher email].

Appendix E: Phase 2 checklist for WISEflow submission

1. Replace all [INSERT] placeholders with final data. 2. Confirm SIKT approval number on consent page. 3. Run plagiarism check (Ouriginal). 4. Filename: Etternavn_Fornavn_studentnummer_ACIT5920. 5. Append survey PDF export as ACIT5920_Appx if required. 6. Verify figure alt-text. 7. Supervisor sign-off obtained.

Interview materials (Phase 3)

Appendix A: Interview guide (draft)

1. Walk me through a time you received a message you thought might be fraudulent.

What did you notice first?

2. What makes a message feel trustworthy or suspicious?

3. How do you handle unexpected emails from IT, exam office, or BankID/Altinn?

4. Have you ever almost clicked something you later realised was suspicious?
What was happening in your life then?

5. Do feelings of loneliness or stress affect how you respond to messages?
(Optional skip)

6. What training have you had? What was missing?

7. If you made a mistake, would you report it? Why or why not?

8. Anything else about digital fraud and student/work life?

Appendix B: Codebook template

Code | Definition | Example excerpt | Theme

institutional_trust | Message perceived as official/expected | "It looked like OsloMet IT" | Theme 1

stranger_dismissal | Quick rejection of unknown sender | "Obviously fake lottery" | Theme 1

belonging_pull | Social inclusion motive | "I wanted to join the group" | Theme 2

exam_pressure | Urgency during exams/deadlines | "Had three deadlines that week" | Theme 2/3

know_but_click | Articulates rules yet nearly acted | "I knew I should check" | Theme 3

shame_silence | Would not report due to embarrassment | "I'd feel stupid" |

Theme 4

supportive_report | Would report if culture supportive | "Better to tell IT" |

Theme 4

Appendix C: Interview consent (outline)

Recording consent; pseudonym use; secure storage; right to withdraw; SIKT reference; welfare contacts.

Appendix D: Final thesis chapter merge checklist

1. Preface and abstract (updated with results). 2. Ch 1-2 from Phase 1 (edited). 3. Ch 3 Methodology (Phase 2 survey + Phase 3 interviews). 4. Ch 4 Results (survey + themes + integration). 5. Ch 5 Discussion. 6. Ch 6 Conclusion. 7. References merged. 8. Appendices: survey, vignettes, interview guide, ethics. 9. List of figures/tables. 10. WISEflow ACIT5930 + Appx zip.

Appendix E: WISEflow submission checklist

Filename Etternavn_Fornavn_studentnummer_ACIT5930; Ouriginal check; all [INSERT] removed; SIKT docs in Appx; figure alt-text; page numbers; 19 May deadline.

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